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REAL-TIME BUS ARRIVAL INFORMATION SYSTEM – AN EMPIRICAL EVALUATION

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Abstract

The provision of real-time information concerning bus arrival times could potentially reduce the uncertainty associated with public transport trips and improve the overall level-of-service. In addition, real-time predictions might enable operators to apply proactive control strategies. Even though considerable research efforts were devoted to the development of bus arrival prediction schemes, there is lack of knowledge on the performance of real-world operational systems. This paper aims to investigate the performance of a commonly deployed real-time information generation scheme. A conventionally used scheme is implemented and evaluated based on an empirical analysis. Performance metrics concerning the prediction error accuracy and reliability and their impact on expected waiting time were formulated from both passengers' and operators' perspective. The real-time information generator was applied on the trunk line network in Stockholm, Sweden. The accuracy and reliability of the prediction scheme was analysed by comparing the generated predictions against vehicle positioning data. This scheme was found to systematically underestimate the remaining waiting time by 6.2% on average. The provision of real-time information yields a waiting time estimate that is more than twice closer to the actual waiting times than the timetable is. This difference in waiting time expectations is equivalent to 30% of the average waiting time.

Keywords: Real-time information; Bus; Prediction scheme; Waiting time: Reliability

1. Introduction

In recent years, Intelligent Transportation Systems (ITS) have been extensively deployed in planning, control and management of modern transportation systems. Public transport agencies have either successfully implemented or are in the process of transforming their systems by adapting ITS applications for planning, maintenance and operational purposes (Ezell, 2010). Advanced Traveller Information Systems (ATIS) constitutes a class of ITS which provides real-time travel and network information regarding public transport routes and schedules as well as information about delays and accidents.

Public transport systems are increasingly equipped with information and communication technologies in order to improve passenger level of service and facilitate fleet management (Casey et al., 2000). The dissemination of accurate and reliable information to passengers is of great importance for public transport agencies. ATIS and Advanced Data Collection (ADC) techniques, such as Automatic Vehicle Location (AVL) and Automatic Passengers Counts (APC), were first used in order to improve operations and management. Later on, they were also utilized to disseminate real-time information (RTI) to passengers (FHA and FTA 2000; TCRP 2008). With the development of such systems, public transport agencies obtain details regarding fleet positioning, travel time and speed, passengers on board and dwell time at stops. These sources are then processed and integrated in order to generate RTI concerning present and future states of the public transport system. RTI could refer to information on service disruptions, crowding conditions and prescriptive journey planners. However, the most commonly provisioned as well as the main focus of research is information concerning the remaining waiting time since it constitutes an important and uncertain part of the trip.

1.1. Information Provision

The two most commonly provisioned sources of information regarding bus arrival times are timetables and RTI services. Static timetables are constructed based on fleet scheduling, aggregated passengers' demand profiles and aggregated traffic conditions. Timetables have thus a limited value as a source of information as they are designed based on historical states of the system which reflect its performance during a specific time period (e.g. data collection and timetable design periods). Moreover, historical travel times are influenced by drivers' attempts to adhere to the timetable. The latter is thus not merely a reflection of travel times but rather an important determinant of these times (Cats 2014). Furthermore, static timetables are inferred from aggregate travel time distributions and hence do not reflect instantaneous variations in service performance. Caulfied and O'Mahony (2009) investigated the opinion of passengers concerning public transport information provision. The provision of timetables at stops received the lowest score among bus passengers. In contrast, passengers' preference towards RTI is supported by numerous empirical studies which reported that 70-100% of all passengers use RTI as a source of information whenever available (TCRP 2003). The key advantage of RTI is its reliance on the current conditions of the network, which can thus yield more accurate predictions of bus arrival times.

Previous studies analysed the impact of RTI provision on various aspects of travellers' experience including the level of satisfaction (Zhang et al. 2008) and perceived waiting time (Mishalani et al. 2006; Dziekan and Kottenhof 2007). Watkins et al. (2011) delved into the importance of providing accurate information and its implications on passengers' perception of waiting time. In their study passengers perceived the waiting time as 0.83 minutes longer than the actual waiting time they have experienced. In contrast, this overestimation of waiting times did not occur when RTI was available. Moreover, the average actual waiting time of the latter group was 30% shorter than waiting time experienced by passengers who have used traditional arrival time information (i.e. timetables).

The benefits of RTI are not limited to a single trip stage. Labell et al. (1992) reported the advantages of RTI in: improving transfer and departure time decisions based on pre-trip information (e.g. initial connections); improving time usage in/near the terminal, and; informing passengers on irregularities in public transport services, and hence reducing passengers anxiety associated with travel time uncertainty. Moreover, the benefits from deploying RTI systems are not limited to uncertainty reduction and trip departure time choice. RTI can also facilitate path choice changes that would yield time savings (Nökel and Wekeck 2009, Cats et al. 2011).

From the operators' perspective, the existence of real-time tracking systems is instrumental in monitoring and managing public transport operations. Predictions concerning the progress of the vehicle fleet allow operators to gain insights into expected future states, compare them with the planned service, and consider alternative measures for improving system performance. In case performance-based incentives are embedded into the agreement between the operator and the public transport agency, effective real-time control might have considerable economic implications for the operator. By monitoring the performance of the system online, the operator is able to react instantaneously or even take a proactive approach in order to mitigate anticipated service irregularities.

1.2. Real-time information generation and evaluation

The generation of RTI requires the collection, integration and processing of instantaneous data, which are then used as an input to prediction algorithms. Data sources may consist of a combination of historical travel times, scheduled arrival times and AVL data for preceding vehicles. The aforementioned algorithms provide the expected vehicle arrival times based on a set of assumptions concerning the system evolution. Previous studies applied various methods for bus arrival predictions including regression models (Jeong and Rilett 2005), artificial neural networks (Mazloumi et al. 2011), Kalman filter (Cathey and Dailey 2003; Shalaby and Farhan 2004), support vector machines (Bin et al. 2006, Yu et al. 2011) and statistical pattern recognition (Vu and Ata 2010). These methods are consistent with the general prescription proposed by Cathey and Dailey (2003).

Even though considerable research efforts were devoted to the development of bus arrival prediction schemes, which involve the application of computationally-intensive statistical methods, there is lack of knowledge on the performance of real-world operational systems. Previous studies that assessed the impacts of RTI provision were based on comparing simulated

RTI to simulated AVL of bus operations (Hickman and Wilson 1995, Mishalani et al. 2000, Nökel and Wekeck 2009, Cats et al. 2011).

Empirical evaluation of real-world RTI services is necessary in order to assess the performance of currently deployed systems and estimating their benefits. The evaluation also enables the identification of RTI system's limitations and conditions under which the system underperforms that should be addressed by future developments. To the best of our knowledge, no previous study conducted an empirical evaluation of a real-world real-time transit information.

1.3. Study approach

This paper aims to investigate the performance of a commonly deployed RTI generation scheme. The scheme is provided by one of the three leading global ITS providers and is thus used by many public agencies worldwide, including the public transport authority in Stockholm, Sweden (SLL). The scheme evaluated in this study was formulated based on direct discussions with the information technology department of the system provider and the technical department of SLL. The system provider confirmed that the scheme evaluated in this paper is currently used by most of its customers.

A conventional timetable-based real-time bus arrival prediction scheme is evaluated based on an empirical analysis. Such an empirical evaluation of RTI generation requires the comparison of actual and predicted vehicle arrival times. However, the predictions are often produced and provisioned by a commercial software provider. Thus, the prediction system is not managed or accessed by public transport operators and/or agencies. Private system providers do not release descriptions of their prediction schemes due to commercial interests. Moreover, historically provisioned information may not be systematically stored by the system provider. In the lack of unmediated access to data concerning RTI provisions, the prediction scheme was formulated and implemented. The real-time information generator was then applied on vehicle positioning data from Stockholm. Thereafter, the generated predictions were compared against the ground-truth bus arrival data.

The rest of the paper is organized as follows. The RTI generation method is formulated in Section 2 along with the respective performance metrics from passengers' and operators' perspectives. Section 3 presents the case study and the results including a comparison of RTI and static timetable information. This paper concludes with an overall assessment of the current system and identifies potential areas for the development of more elaborative prediction schemes.

2. Method

2.1. Real-Time Information Generation Scheme

The bus arrival prediction scheme evaluated in this study is based on instantaneous data concerning the position of the approaching bus and the corresponding remaining scheduled travel time. It requires therefore the real-time positions of all buses and a time-dependent timetable

database. The latter is seasonally constructed and indicates planned arrival times at each stop along the line. Many bus services use a subset of the stops for regulating departure times according to the timetable. Drivers are instructed to hold at these stops in case they run early compared with the timetable in order to improve service punctuality. The dispatching from the origin stop at the terminal is typically regulated based on the timetable. The set of regulation stops, also known as time point stops (TPS), has to be indicated in the timetable.

The commonly used scheme that is evaluated in this paper is based on few simple prediction rules for generating RTI. The prognosis scheme is based on the following assumptions:

(a) The travel time between bus current location and any downstream location is equal to the scheduled travel time;

(b) Buses never leave TPS prior to their scheduled time.

The combination of these assumptions implies that buses maintain their schedule deviation – earliness or lateness. The only exception is buses that run early and have a TPS between the current location and the relevant downstream location, since they will be able to correct their schedule deviation and will therefore arrive on-time. Conclusively, a bus that runs behind schedule is always expected to maintain its current delay at all downstream stops.

Bus trajectory could be represented as a vector of time stamps along a list of locations, typically stops. The trajectories of an ordered set of bus trips denoted K_l on line $l \in L$ during a certain time interval can be hence represented as a matrix, where L is the set of service lines in the network. Let us denote this matrix as π^a where each cell, $\pi^a_{k,s}$, is the actual time where bus trip k arrived at location $s \in S_l$. This matrix is partially empty at time instance τ for any ongoing trip. A subset of the recording locations for line l ($\hat{S}_l \subseteq S_l$) serves as TPS. A corresponding matrix denoted π^t contains the timetable trajectories for K_l. The output of the prediction scheme is the corresponding matrix of predicted bus arrivals, π^p.

The two fundamental assumptions of the prognosis scheme could be then formulated as:

$$\pi^a_{k,j} - \pi^a_{k,i} = \pi^t_{k,j} - \pi^t_{k,i} \;\; \forall i,j \in S_l, i \prec j \tag{1}$$

$$\pi^a_{k,j} = \pi^t_{k,j} \;\; \forall j \in \hat{S}_l \tag{2}$$

Equations 1 and 2 are the corresponding mathematical expressions of the abovementioned assumptions (a) and (b), respectively.

Figure 1 presents the process of generating RTI for the remaining time until the next vehicle arrival on line l at stop s. The prediction scheme consists of the following steps when generating at time τ the prediction of the next arrival of line l at stop s, $\pi^p_{l,s}(\tau)$:

(a) *Find the last bus trip k that visited stop s* – Let us denote by $k^p \in K_l$ the latest bus trip that visited stop s, hence:

$$k^p = arg\ max_{K_l}\{\pi^a_{k,s} : \pi^a_{k,s} < \tau\} \tag{3}$$

(b) *Find the last location visited by the next bus trip* - Let us denote by m the last location that was visited by the next trip $(k^p + 1)$, defined as follows:

$$m = \begin{cases} if\ \pi^a_{k^p+1,1} \neq \emptyset & arg\ max_{i=1\ldots s}\{\pi^a_{k^p+1,i} : \pi^a_{k^p+1,i} < \tau\} \\ Otherwise & 0 \end{cases} \quad (4)$$

The second condition ensures that if the next trip has not started yet ($\pi^a_{k^p+1,1} = \emptyset$), m will point to the dispatching terminal.

(c) *Make a prediction based on the timetable* – The scheme distinguishes between the two following cases:

I. At time τ, the next trip to visit stop s has not started yet ($m = 0$) OR the bus is running early ($\pi^a_{k^p+1,m} < \pi^t_{k^p+1,m}$) and there is an intermediate TPS ($\exists\ m \leq i < s, i \in \hat{S}_l$) then the predicted arrival time is simply the scheduled time:

$$\pi^p_{l,s}(\tau) = \pi^p_{k^p+1,s}(\tau) = \pi^t_{k^p+1,s} \quad (5)$$

II. Otherwise ($\pi^a_{k^p+1,m} \geq \pi^t_{k^p+1,m}$OR $\nexists\ m \leq i < s, i \in \hat{S}_l$), the predicted arrival time is calculated based on the scheduled remaining travel time:

$$\pi^p_{l,s}(\tau) = \pi^p_{k^p+1,s}(\tau) = \pi^a_{k^p+1,m} + \pi^t_{k^p+1,s} - \pi^t_{k^p+1,m} \quad (6)$$

In other words, it is assumed that the current deviation from the scheduled will be sustained in case of non-early trips as well as in case there is no intermediate TPS.

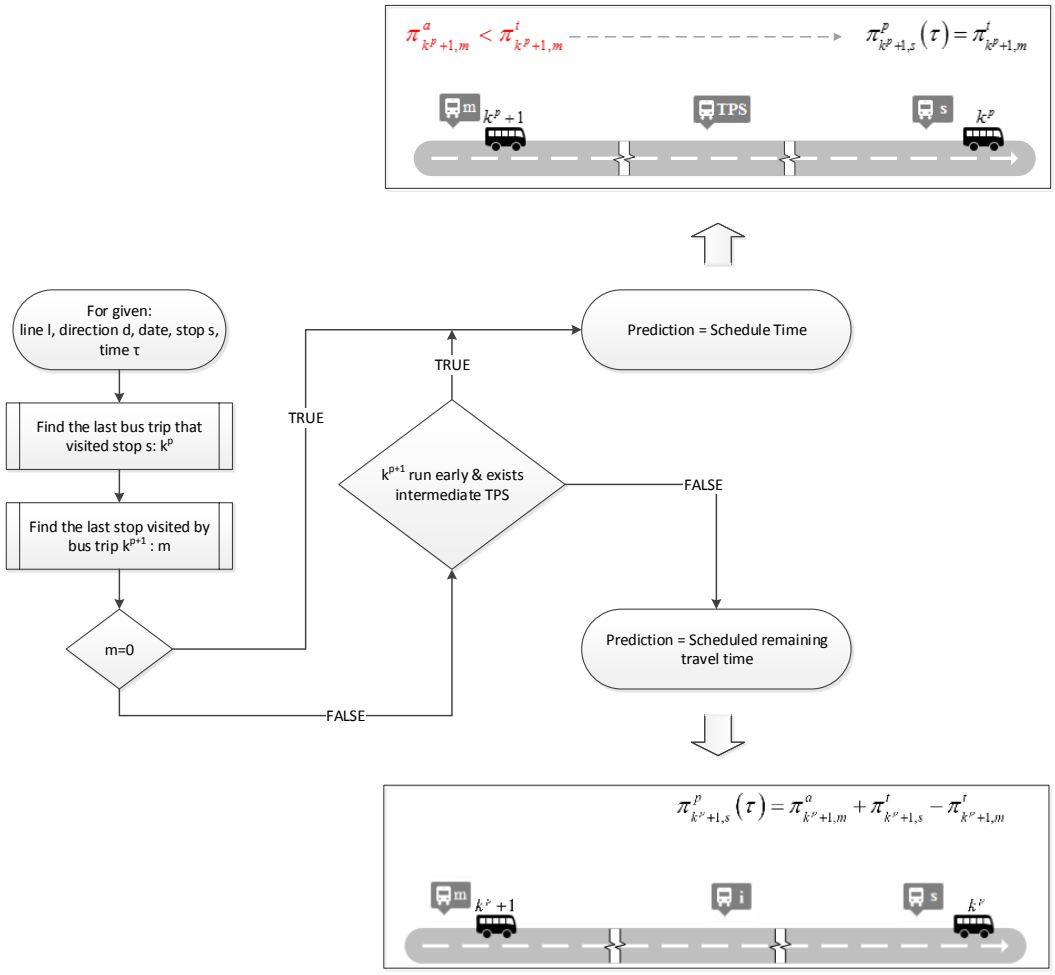


Figure 1: Prediction scheme flowchart

2.2. Performance metrics

The RTI performance is assessed by a series of metrics that are calculated ex-post and consider both passengers' and operators' perspectives. While the former is concerned with the arrival of any vehicle at a certain line, the latter is interested in arrival times of specific vehicles and the resulting trajectory. The prediction error for the arrival of trip k at stop s is therefore assessed by comparing the prognosis generated at time τ against the corresponding actual arrival time of the same trip:

$$e^{p}_{k,s}(\tau) = \pi^{a}_{k,s} - \pi^{p}_{k,s}(\tau) \qquad (7)$$

From passengers' perspective, however, no importance is attached to the specific trip's identity, and the accuracy is determined by the difference between the provisioned RTI and the next arrival of line l at stop s, calculated as follows:

$$e^{p}_{l,s}(\tau) = \pi^{a}_{k^{a},s}(\tau) - \pi^{p}_{l,s}(\tau) \qquad (8)$$

Where k^a is the first trip to arrive at the stop, defined as:

$$k^a = arg\ min_{K_l}\{\pi^a_{k,s} : \pi^a_{k,s} > \tau\} \quad (9)$$

This could be interpreted as the difference between the predicted and experienced waiting times for a passenger that arrived at stop s at time τ. Note that k^a might differ from k^p when an overtaking occurs between m and s.

The prediction error measures enable to identify the difference between predicted and observed arrival times. Moreover, the variability of prediction errors has to be minimized in order to obtain a reliable prediction scheme.

The performance of static information concerning arrivals is used as a benchmark for assessing the added-value yielded from RTI provision. Static information accuracy from operators' and passenger's perspectives - $e^t_{k,s}$ and $e^t_{l,s}(\tau)$, respectively - was formulated similarly by substituting the RTI prediction for the corresponding timetable term, as follows:

$$e^t_{k,s} = \pi^a_{k,s} - \pi^t_{k,s} \quad (10)$$

$$e^t_{l,s}(\tau) = \pi^a_{k^a,s}(\tau) - \pi^t_{k^t,s}(\tau) \quad (11)$$

Where k^t is the first trip scheduled to arrive at the stop, defined as:

$$k^t = arg\ min_{K_l}\{\pi^t_{k,s} : \pi^t_{k,s} > \tau\} \quad (12)$$

Note that k^a might differ from k^t in case the first arriving bus was scheduled to arrive before the passenger arrival time at the stop. The prediction error of static information from operators' perspective (Eq. 10) is equivalent to schedule adherence at the vehicle-level.

Furthermore, the extent to which timetables and RTI are effective in assisting passengers to shift their expectations closer to the actual waiting time is assessed. The actual waiting time of a passenger arriving at stop s at time τ with the intention to board line l is:

$$w^a_{l,s}(\tau) = \pi^a_{k^a,s}(\tau) - \tau \quad (13)$$

While the expected waiting time implied by the timetable and RTI are, respectively:

$$w^t_{l,s}(\tau) = \pi^t_{k^t,s}(\tau) - \tau \quad (14)$$

$$w^p_{l,s}(\tau) = \pi^p_{l,s}(\tau) - \tau \quad (15)$$

The mean absolute error provides an aggregate measure of performance which could be compared across lines, systems, time periods and prediction scheme. It is calculated as follows:

$$MAE^t = \overline{\left|w^a_{l,s} - w^t_{l,s}\right|} = \frac{\sum_{l,s,\tau}|e^t_{l,s}(\tau)|}{\sum_{l\in L}|S_l|\cdot|\tau|} \quad (16)$$

$$MAE^p = \overline{\left|w_{l,s}^a - w_{l,s}^p\right|} = \frac{\sum_{l,s,\tau}\left|e_{l,s}^p(\tau)\right|}{\sum_{l\in L}|S_l|\cdot|\tau|} \quad (17)$$

Where $|S_l|$ and $|\tau|$ denote the number of stops on line l and the number of RTI generation instances considered in the analysis (e.g. analysing half a minute generation-intervals for the peak hour will result with 120 instances), respectively. This measure corresponds to the average time difference obtained by a certain information source when compared with the observed bus arrival times.

3. Application

3.1. Case Study

3.1.1. Vehicle Positioning Data

The entire bus fleet in Stockholm is equipped with an AVL system. Buses communicate their location every 15 seconds as long as the bus is on duty. The system is used for several purposes including radio communication, real-time monitoring and control of vehicles, fleet management strategies and the generation of real-time passenger information. Moreover, the real-time vehicle location probes are automatically processed for computing service performance indicators such as punctuality and efficiency. The database used for analysis purposes consists of a vehicle position record for each bus stop and bus trip combination. This event-based database is constructed by computing the arrival and departure time of each vehicle trip from each stop along its route. Departure and arrival times are computed by the AVL system provider based on the bus location and its proximity to a stop along the relevant route, route sequence, dwelling time and doors opening and closing times. Each stop visit record contains information regarding the trip ID, stop ID, actual and scheduled arrival times as well as actual and scheduled departure times.

The performance of the RTI generation method was analysed based on detailed and comprehensive AVL data. These data were provided by the regional public transport authority, SLL, and contained the abovementioned vehicle position data for each bus stop. The study period consists of records from 15/11/2011-15/12/2011 and 9/1/2012-19/1/2012, in order to exclude the winter holidays. Only records between 07:00-19:00 were considered for further analysis resulting with a dataset of more than one million records.

The trunk line network of Stockholm was selected as the case study network. As shown in Figure 2, it includes four lines which compose the backbone of Stockholm inner-city bus network. These lines are characterized by high frequency, articulated vehicles, designated lines on main streets, traffic signal priority and RTI displays at all stops. The case study network contains more than 200 stops along more than 80 route-km. Each route includes 2-4 time point stops (TPS) located at key public transport transfer locations. Each bus in Stockholm is equipped with a computer called BussPC located in the driver's cabin which constantly displays the current schedule deviation, allowing drivers' to monitor their punctuality and adjust their speed accordingly. These lines account for 60% of the total ridership in this area with approximately 120,000 boarding passengers per day between 7:00-19:00 (SL 2006).



Figure 2: Stockholm's inner-city trunk line routes

3.1.2. Implementation Details

Real-time bus arrival predictions were generated based on the method described in Section 2. The raw vehicle positioning data (e.g. actual arrival) and the corresponding timetable were first transformed into a matrix format resulting with π^{α}, and π^{t}, respectively. Concerning the construction of the matrix, the bus positioning data were ordered based on the sequence of scheduled trips throughout the analysis period. Note that the data contain also partial trips. Such trips cover only a subset of the stops along the respective line, as the operations of some lines consist of several trip patterns. The prediction scheme handled this explicitly by ensuring that the arrival prediction refers to a bus that is destined to serve the respective stop. In case of missing records, arrival times were interpolated. Since the prediction scheme is passenger-oriented, the provided information is based on the approaching vehicle rather than the next scheduled vehicle. With the ability to detect multiple bus trajectories, overtaking is taken explicitly into account. In other words, k^{p} may refer to different trips when generating predictions at stop s on successive time instances with no intermediate vehicle arrival.

The input data – vehicle positioning and timetables - were available at the stop-visit level, while the prediction outputs were generated every minute. In other words, the event-based data availability was used for generating instantaneous predictions in a time-based fashion. The output of the prediction scheme provides a snapshot of the countdown RTI provisioned at every

stop across the study network with one minute interval throughout the study period. The RTI generator results with the predicted bus arrival time at each stop and line combination, $\pi_{l,s}^{p}(\tau)$, for every minute so that $\tau = (7{:}00, 7{:}01, \ldots, 19{:}00)$. The prediction scheme was implemented in MATLAB. The RTI generator at time τ follows then the steps outlined in Section 2 and thus utilizes only bus positioning data collected prior to time τ.

The output produced by the implemented RTI generator enabled the computation of the performance metrics defined above. The performance of the RTI generator was analysed by comparing predicted and actual arrival times. The fundamental analysis unit consists of a cross-network sampling of the prediction provided by the RTI systems. This implies the calculation of the passengers' and operators' prediction error metrics across the network with one minute sampling. The implementation facilitates the evaluation of the RTI generator across time and space and the potential implications on passenger waiting times. Furthermore, "SL minute", named after the public transport agency, is notoriously known and used by the public and popular media in Stockholm as a particularly "long" minute because the bus fails to arrive within the projected time window. The design of this study enables to examine whether the coined term is empirically justified for the current system.

3.2. Analysis and Results

3.2.1. Prediction accuracy

In order to evaluate the overall accuracy of the bus arrival prediction scheme (Section 2.1), the distribution of the prediction error was constructed (Eq. 8) and is presented in Figure 3. The prediction error corresponds to the difference between the actual and predicted arrival times computed for each time instance τ during the study period and across all stops included in the case study network. The prediction accuracy follows a normal distribution with a mean value of +15 seconds and a standard deviation of 2 minutes. Moreover, the distribution is positively skewed, indicating that the prediction scheme has a slight systematic bias resulting in an underestimation of passengers' waiting time. More than one third of the inquiries (36%) yielded a prediction error - either positive or negative - of more than 1 minute. Furthermore, 14% and 5% of RTI projections had a prediction error of more than 2 and 4 minutes, respectively. In two thirds of these cases, the prediction error was due to an underestimation of the remaining waiting time.

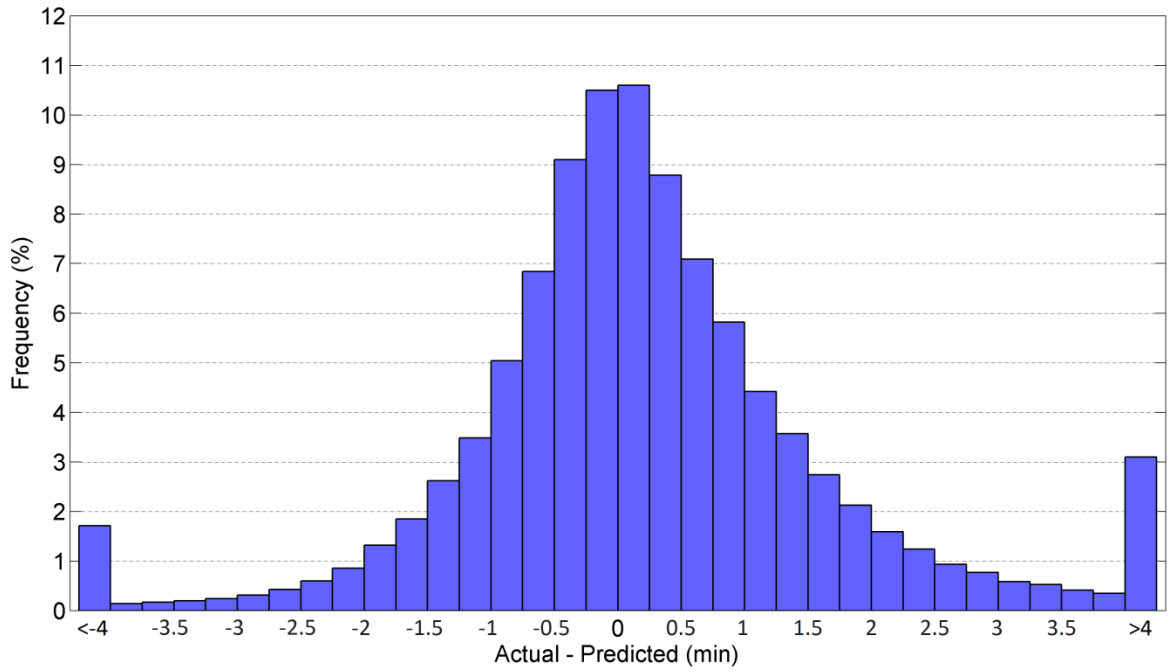


Figure 3: Real-time information prediction error

The analysis of the prediction accuracy allows quantifying the extent to which the notorious abovementioned "SL minute" manifests itself given the currently implemented RTI prediction scheme. The prediction error of the provisioned RTI should be assessed by accounting for the average waiting time. This analysis obtained that the average excessive waiting time per projected minute is 6.2% for the case study system. In absolute terms, the average "SL minute" lasts in fact 63.7 seconds. However, the average excess waiting time disguises substantial variations which are explored in the following sections.

3.2.2. Prediction horizon

The accuracy and the reliability of the prediction scheme depend greatly on the prediction horizon as expressed by the time lag between the RTI generation time and the next bus arrival time. As could be expected, prediction accuracy and reliability improves for shorter prediction horizons. In other words, RTI performs better when the remaining waiting time is shorter. The bars in Figure 4 correspond to the relative difference between the actual waiting time (Eq. 13) and the waiting time projected by RTI (Eq. 15) for different actual waiting times. For very short waiting times the prediction scheme overestimates the remaining waiting time. In contrast, an increasingly underestimation is observed for waiting times longer than 2 minutes. Particularly, for waiting times longer than 8 minutes, the underestimation reaches almost 10%. For example, a RTI projection of 9 minutes would on average imply that the waiting time would last almost 10 minutes.

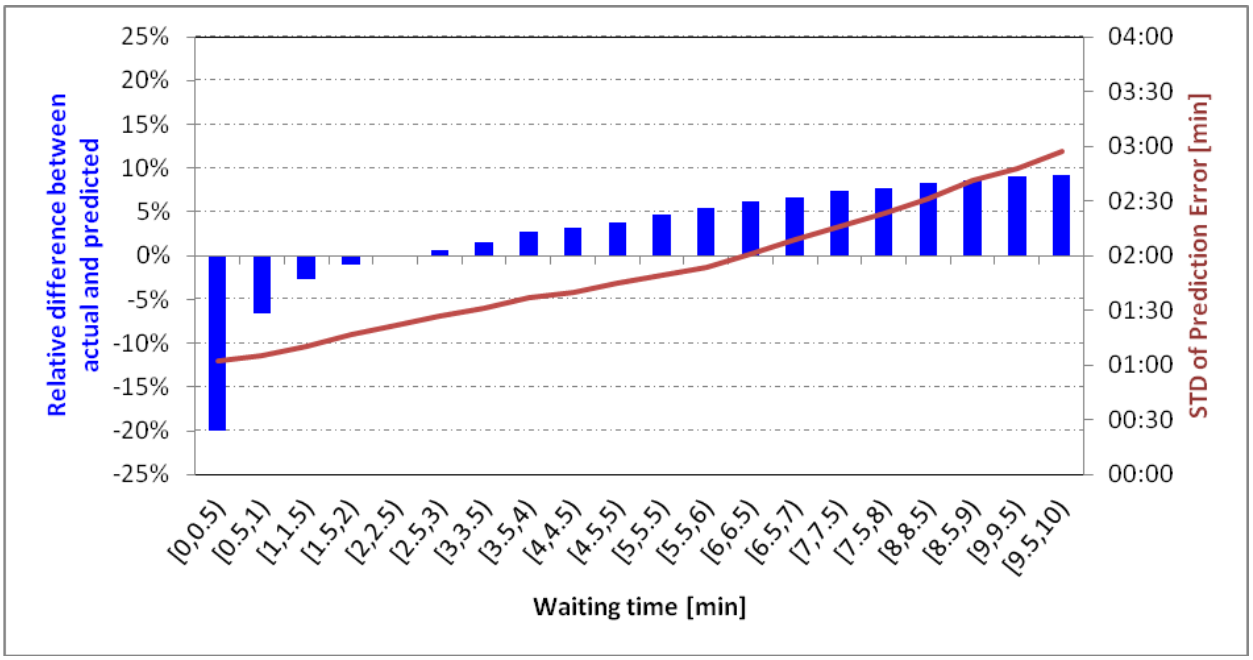


Figure 4: Real-time information accuracy and reliability as function of the remaining time until the next bus arrival

In parallel, the reliability of RTI prediction improves monotonically as the prediction horizon becomes shorter. The prediction reliability measured in terms of the standard deviation of the prediction accuracy is presented in the above figure as a function of the remaining waiting time. The curve indicates that the RTI prognosis is less reliable for longer prediction horizon. This was expected as longer prediction horizons undermine the possibility to estimate the sequential time disturbances. Since the standard deviation of RTI prediction error increases linearly with the waiting time, the reliability does not increase in relative terms (e.g. coefficient of variation).

3.2.3. Temporal Analysis

The temporal variations of traffic and passenger demand conditions result with systematic variations in RTI prediction accuracy and reliability with respect to the day of the week and time of the day. Table 1 shows the mean and standard deviation values of the prediction error by day of the week. While the underestimation of waiting times as manifested through positive prediction errors prevails on all days of the week, its extent varies considerably. On Mondays and Tuesdays the RTI predictions were most accurate and reliable. However, the prediction accuracy and reliability deteriorate along the week with the worst performance obtained on Saturdays. Sundays, in contrast, performed like an average weekday. These differences are presumably determined by the extent to which timetables reflect the prevailing traffic conditions as the timetable underlies the prediction scheme. The case study network has a common timetable for all weekdays and separate timetables for Saturdays and Sundays.

Table 1: Prediction Error by Day-of-the-Week (mm:ss)

	MON	**TUE**	**WED**	**THU**	**FRI**	**SAT**	**SUN**
Mean	00:05	00:10	00:17	00:21	00:22	00:32	00:20
StD	01:51	01:46	02:18	02:25	02:38	02:52	01:50

A further analysis regarding the temporal variations with respect to time of day was conducted. Particularly, prediction errors were calculated separately for trips that started at the AM peak (7:00-9:00), Off-Peak (9:00-15:30) and PM peak (15:30-19:00) periods. The latter is associated with less accurate and less reliable RTI provision as evident in Table 2. In contrast, the AM peak performs surprisingly well. This may be due to its more homogeneous traffic and demand conditions.

Table 2: Prediction Error for Time-of-Day Categories (mm:ss)

	Time-of-day periods		
	AM	***Off-Peak***	***PM***
Mean	00:11	00:10	00:15
St.D.	01:35	01:49	02:12

3.2.4. Spatial Analysis

Service reliability tends to deteriorate along the route both in terms of punctuality and regularity (Ji et al., 2010; Cats et al., 2012). The performance of the prediction scheme was therefore also analysed along each of the case study routes in order to identify spatial variations. The eastbound route of bus line 1 was selected to illustrate the commonly observed patterns. Figure 5 illustrates the performance of the prediction method along this route. It consists of 33 stops, of which three serve as TPS marked with stars (Fridhemsplan, Hötorget and Värtavägen). Specifically, the blue curve follows the performance of mean across different stops. At the same time the vertical lines show the relative standard error. It can be observed that the average prediction error (Eq. 8) fluctuates along the route within the range of ±1.5 minutes.

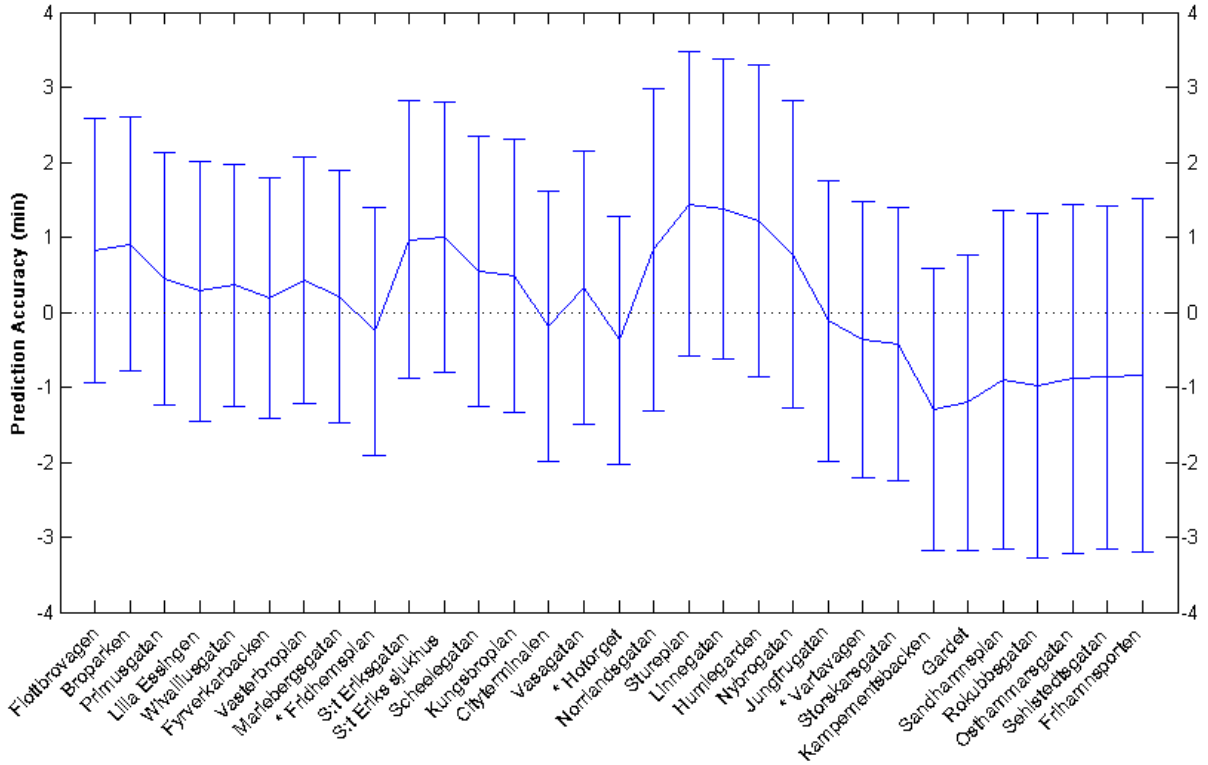


Figure 5: Real-time information accuracy and reliability along the route, Line 1 eastbound; (*) indicates a time point stop

Furthermore, the analysis of the performance along the route revealed the following patterns:

I. RTI predictions become less accurate as well as less reliable at further downstream stops. This could be explained by the propagation of uncertainty associated with traffic conditions, dwell time and driver behaviour along the route. The growing systematic offset from the schedule is not captured by the timetable, which constitutes the core input for the prediction scheme.

II. The important role that TPS plays in the prediction method is apparent in evolution of RTI accuracy along the route. It is clearly evident that RTI provides the most accurate prediction at TPS locations followed by an immediate substantial increase in the prediction error, which is then reduced gradually until reaching the next TPS. The accurate prediction at TPS is presumably obtained due to drivers' attempts to adhere to the schedule at these locations where their punctuality is measured (Cats 2014). In contrast, bus arrivals at stops immediately downstream of TPS are subject to large dwell time variations at TPS due to large passenger flows, holding times and driver shift changes.

III. The prediction error is negative on the last stretch of the route following the last TPS. In line with previous studies (Cats et al. 2012), drivers speed up on the last stretch of stops in order to have a longer break between consecutive trips on their driving duty. This pattern results in negative prediction errors as vehicles arrive earlier than the RTI

generation suggests. This discrepancy highlights thus that the currently deployed prediction scheme evaluated in this paper is not calibrated against an up-to-date local AVL data.

3.3. Real-time information added-value

3.3.1. Effects on waiting time uncertainty

The dissemination of RTI concerning arrival times aims ultimately to reduce the uncertainty associated with waiting time for public transport services. Its availability enables passengers to adapt their travel choices accordingly. Notwithstanding, a perfectly punctual public transport system would hypothetically make RTI provision redundant as static information will suffice. The potential added-value of RTI arises from its utilization of recently transmitted vehicle positioning probes as opposed to static information.

The empirical evidence for the added-value of RTI was investigated by contrasting its predictions with the corresponding waiting time predictions that could be derived from the timetable. The performance metrics presented in Section 2.2 were employed for this purpose. Firstly, the performance of the static timetable which acts as a benchmark in this analysis was investigated by constructing the distribution of schedule adherence (Eq. 10), shown in Figure 6. The distribution is strongly skewed towards late arrivals with a long right tail which corresponds to extreme delays. Only 10% of all arrivals were within an interval of ±15 seconds compared to the timetable.

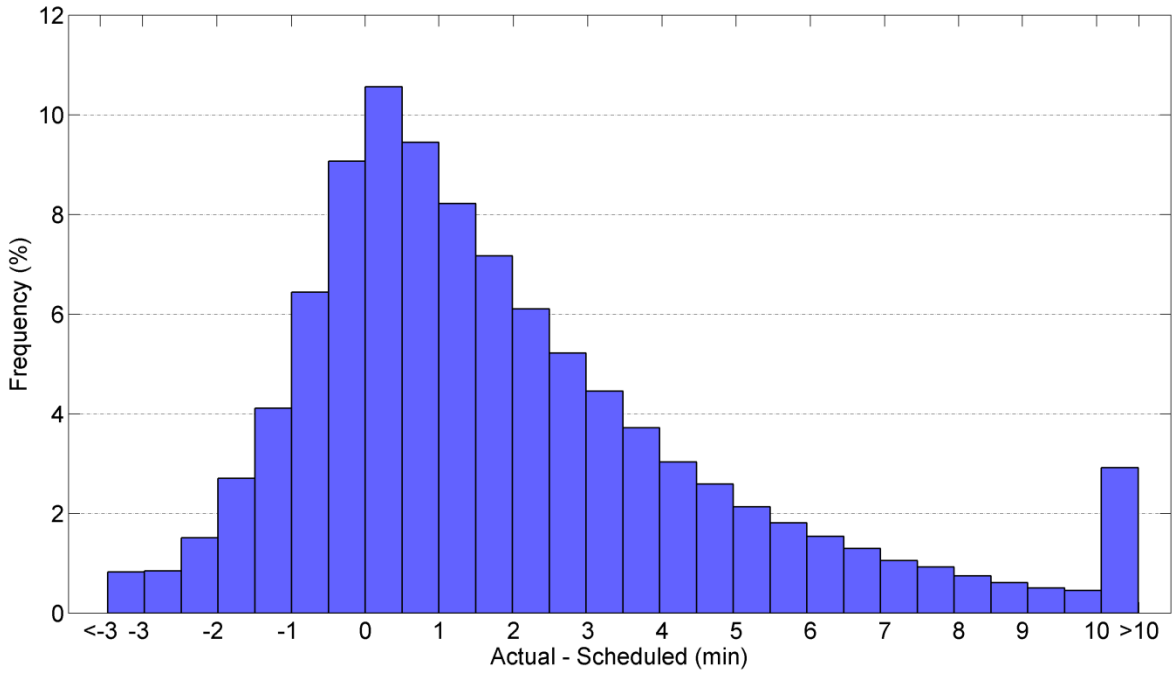


Figure 6: Static information accuracy (vehicle level)

Secondly, the extent to which RTI provisions differ from static timetables was investigated by subtracting the corresponding predictions (Eq. 14-15). As evident in Figure 7, in 20% of the cases real-time and static means yielded the exact same information with respect to the remaining time to the next bus arrival. This could be either attributed to a punctual service or to deficiencies of the prediction method. Given the schedule deviation distribution, a large share of the RTI predictions coincide with those derived from the timetable simply because the prediction scheme reserves to the schedule under certain circumstances. This occurs when the next bus has not started yet its trip or in cases where the bus runs early and there is an intermediate TPS.

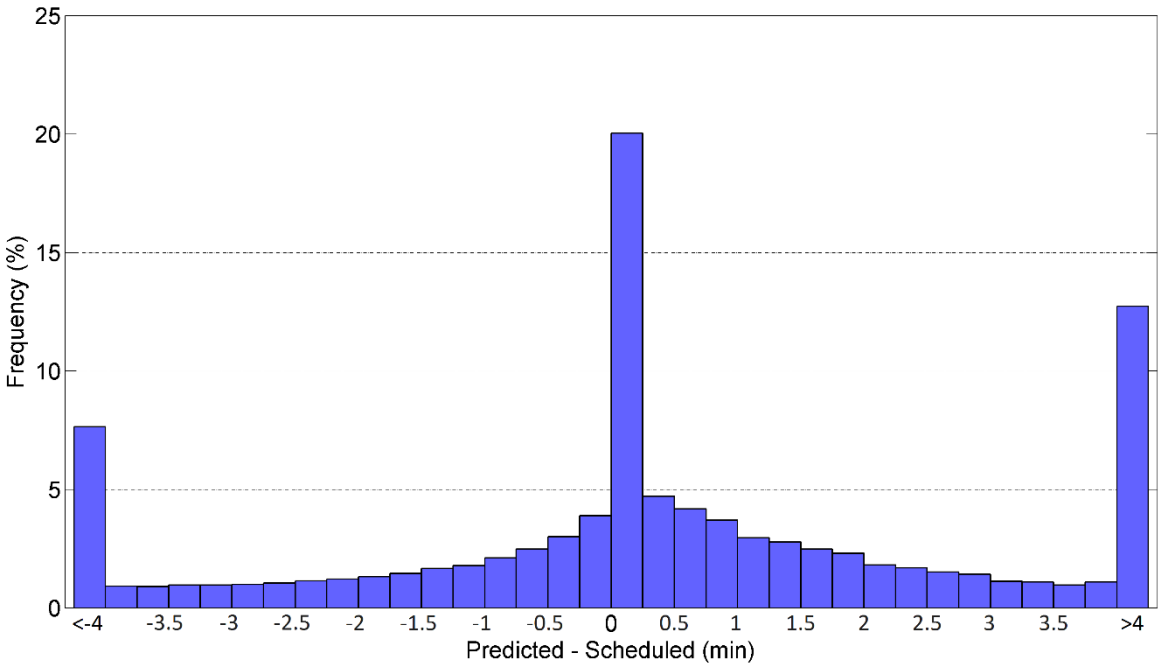


Figure 7: Real-time information vs. static information, passengers' perspective

Thirdly, in order to investigate how RTI effects waiting time uncertainty, the distributions of expected waiting time based on the timetable and RTI (Eq. 14-15) were constructed. Figure 8 contrasts these two distributions as well as compares them with the distribution of the actual waiting times (Eq. 13). The average waiting time, which is the mean actual waiting time derived from the study network during the whole study period, is 4 minutes and 12 seconds with 80% of the passengers waiting less than 5 minutes. The average actual waiting time is 57% longer than the value that would have been obtained from a perfectly regular bus arrival, highlighting the uncertainty involved in bus operations. Note that since the RTI generation procedure followed a uniform temporal generation rate, the number of observations is linearly proportional to the headway.

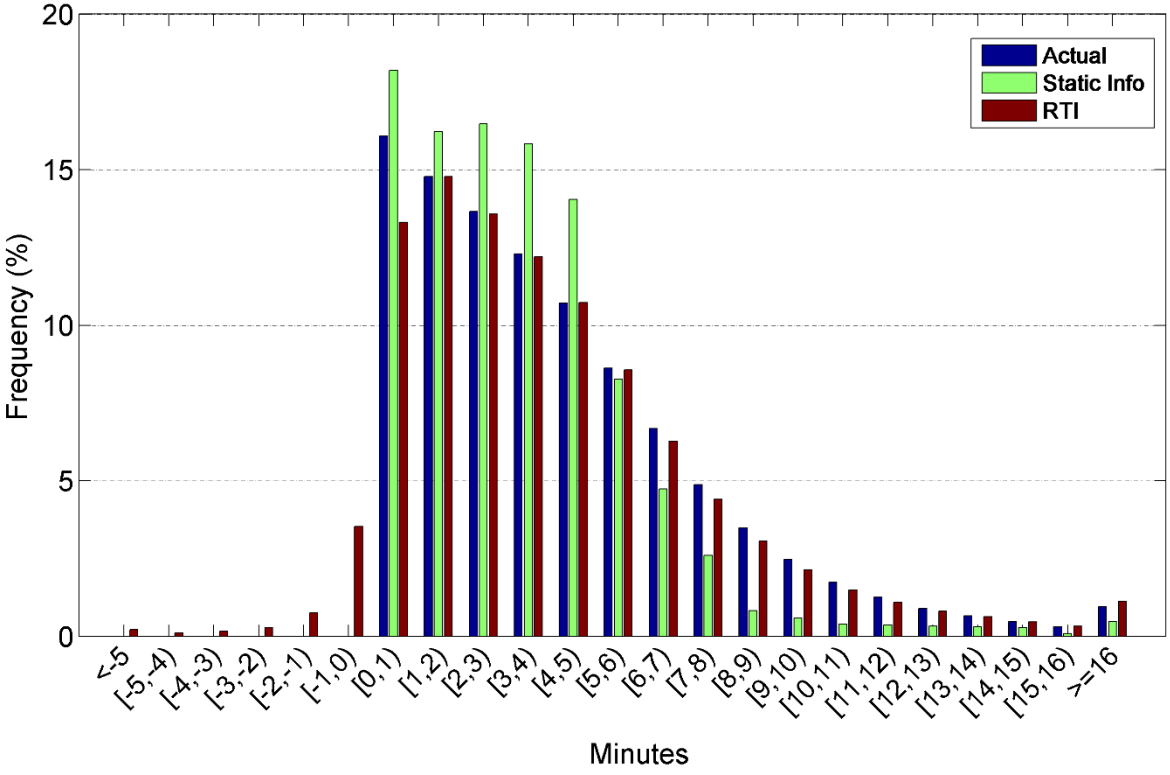


Figure 8: Waiting time distributions - actual and expected based on static or real-time information

It is evident that waiting time expectations based on the timetable result in a considerable underestimation. In particular, a comparison of static information and actual arrivals reveals an overestimation of the likelihood of waiting times shorter than 5 minutes. This could be explained by the fact that the expected waiting time based on the timetable could be realized only in case the service is perfectly punctual.

In contrast, the distribution of expected waiting time based on RTI follows closely the distribution of actual waiting time. The prediction produced by the RTI generation method perform better than the timetable in predicting actual waiting times and hence contribute to a reduction in waiting time uncertainty, an important level-of-service determinant.

The mean absolute error performance measures (Eq. 16-17) enable to quantify the added-value of RTI in reducing waiting time uncertainty. The mean absolute errors of the timetable and RTI are $MAE^t = 146\,sec$ and $MAE^p = 68\,sec$, respectively. This implies that the mean absolute error of passengers' expectations based on RTI is 53% lower than those derived from the timetable. In other words, the prediction scheme yields a waiting time estimate that is more than twice closer to the actual waiting times than the timetable is. This implies that RTI enables passengers to shift their expectations considerably closer to their experienced waiting time. The difference between waiting time expectations derived from the timetable versus RTI is

equivalent to more than 30% of the average waiting time. Notwithstanding, approximately 5% of RTI provision cases yield negative values which correspond to cases where the stop sign displays that the next bus should arrive "Now" while it has not reached the stop yet. It is presumed that this phenomenon contributes significantly to the "SL minute" reputation.

3.3.2. Implications for control and operations

The added-value of RTI is not limited only to implications on passengers' waiting time uncertainty. RTI prediction schemes are also used by operators to project, monitor and ultimately control the progress of their fleet. The information produced by these schemes could project the downstream vehicle trajectory and thus support the decisions taken by control centre dispatchers. The deviation between RTI predictions (Eq. 7) and the timetable (Eq. 10) at the vehicle-level is presented in Figure 9. The added-value of RTI over the planned timetable is greater for operators than for passengers as reflected in the difference between Figure 7 and Figure 9. This difference arises from passengers' indifference to whether a certain bus is the earlier bus running late or the later bus arriving early. Whereas, bus operators are not only interested in the inter-arrival distribution but also in the order of occurrences and how well do they match the planned vehicle scheduling. The comparison between the two perspectives reveals that from operators' perspective there is a significant concentration of observations in the positive tail of the distribution. The arrival predictions are thereof systematically longer than those expected based on the timetable.

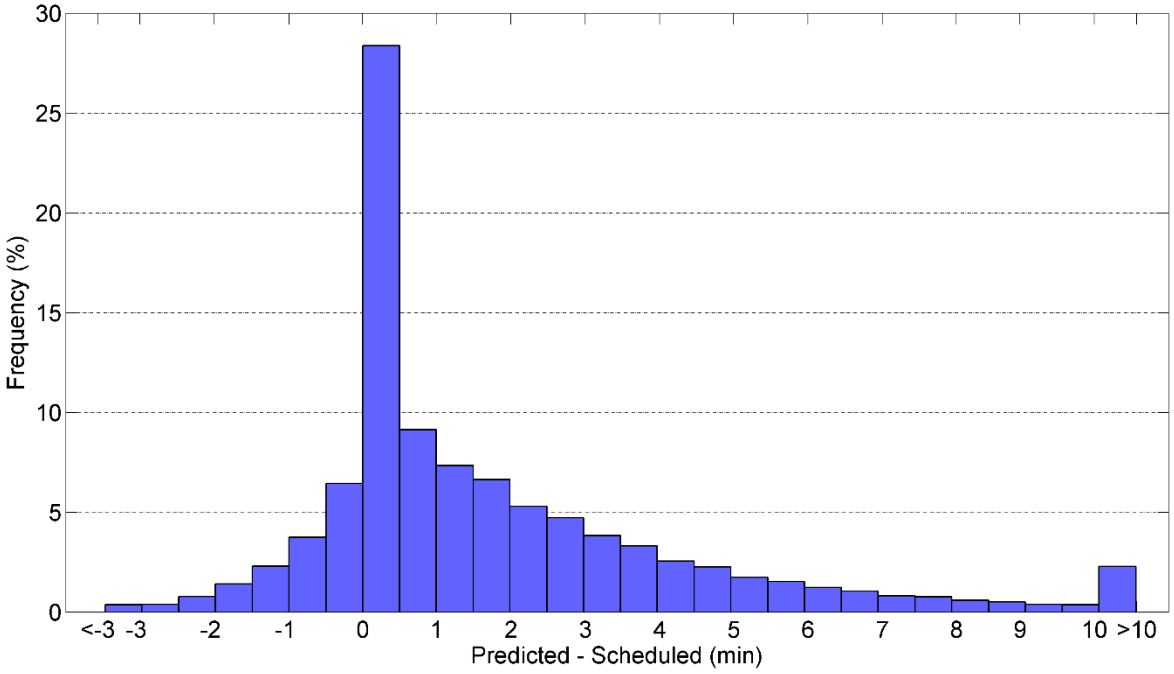


Figure 9: Real-time information vs static information, operators' perspective

4. Discussion

The results presented in the previous section provide an empirical evaluation of the commonly deployed scheme for generating real-time bus arrival information. As evident from the formulation of the scheme in Section 2, the underlying prognosis assumptions are very simplistic. In particular, the prediction logic implicitly assumes the following:

- Travel times required to traverse downstream segments correspond to the scheduled travel times
- Travel times required to traverse downstream segments are independent of the current vehicle performance
- Perfect driver compliance with the control strategy and scheduled dispatching times

Empirical evidence suggests that these assumptions are unrealistic under a large range of circumstances. First, scheduled travel times are not good predictors of downstream travel times in case of considerable day-to-day variability (Mazloumi et al. 2010), travel time variations within the timetable design period and its imperfection (e.g. during the peak hour), traffic incidents (e.g. accident, major demand-generating event). Second, the assumption that future travel times are independent of the bus state variables was empirically disproved by studies that concluded that drivers adjust their speed in response to their schedule deviation (Cats et al. 2012) and have distinguished driving characteristics (Lin and Bertini 2004). Moreover, the positive feedback loop between headways among consecutive buses, number of boarding passengers and dwell times suggest that downstream travel times depend also on bus position relative to other buses. Third, the assumption on perfect driver compliance was empirically tested and found grossly implausible. A lower-bound of the non-compliance rate, which was analysed based on departure times from TPS along line 1, indicates that drivers' compliance with TPS regulation is far from universal. The 37% of the buses that arrived at least 1 minute prior to the scheduled time also departed more than 1 minute ahead of the schedule. Notwithstanding, TPS still play a role in schedule regulation as the corresponding rate for non-TPS was 70%.

The RTI prognosis was found particularly unreliable under the following circumstances: the longer the prediction horizon, on Thursday-Saturday, during the afternoon peak period, further downstream along the route and immediately following a TPS. This highlights the importance of incorporating variations in traffic conditions, passenger activity and driver behaviour in future developments of prognosis schemes. Previous studies have considered a wide range of statistical tools that can obtain more accurate and reliable predictions by using recent vehicle positioning data for calculating the remaining travel time to downstream stops rather than relying on the timetable. Embedding instantaneous AVL data into path-dependent models of trajectory evolution could potentially further improve RTI and its benefits for both passengers and operators.

5. Conclusion

The provision of RTI has the potential to reduce waiting time uncertainty and facilitate proactive control and operations strategies. The performance of a prediction scheme could be assessed by comparing generated arrival predictions against actual arrival times. This paper reports the formulation, implementation and evaluation of a commonly used timetable-based prediction scheme. This scheme utilizes real-time vehicle positioning data concerning only the next

approaching vehicle while the remaining travel time is calculated based on a time-dependent timetable. Performance metrics concerning the prediction error accuracy and reliability and their impact on expected waiting time were formulated from both passengers' and operators' perspective. Equivalent measures were developed for the timetable in order to facilitate the investigation of the added-value induced by RTI.

The performance of RTI provision was applied for the trunk line network in Stockholm's inner-city. The results of this analysis indicate that RTI underestimates the remaining waiting time by 6.2% on average. Given the restricting and naïve assumptions that underlie the prediction scheme assessed in this study, the performance of the RTI system - 64% of all predictions are within ± 1 minute error interval - is considered reasonable by the authors. The performance of RTI was further evaluated by comparing its projections with the respective expectations that could be derived from the static timetable. It was found that the difference between passengers' waiting time expectations derived from the timetable and RTI is equivalent to 30% of the average waiting time. The analysis of RTI performance can contribute to the economic assessment of installing and developing RTI systems. The evaluation can support decision makers prioritizing locations and attributes of the displayed information be assessing the potential waiting time savings. In addition, efforts to improve and enhance prognosis accuracy and reliability could be evaluated by comparing with the benchmarked scheme that was evaluated in this study.

The added-value of RTI was particularly pronounced for operators since vehicle trajectory predictions utilize instantaneous schedule deviation data at the individual vehicle-level. The improvement of predictions scheme is therefore expected to be especially beneficial to bus operators. Real-time predictions enable operators to move beyond reactive monitoring strategies to active and even proactive control, facilitating the deployment of operational measures such as fleet management strategies (Cats et al. 2012, Munoz et al. 2011).

In the absence of records on disseminated RTI, the analysis consisted on generating RTI projections by mimicking the prediction scheme. This implied an event-based vehicle positioning data availability – in terms of stop visit records - rather than time-based – with uniform temporal frequency. In case that the penetration rate of vehicle positioning data and the dissemination from the RTI generator to stop displays is more frequent, our analysis will result in an underestimation of the RTI performance. This is indeed the case in Stockholm where a bus generates three times as many probes as stop-visit records along an average trip.

The deficiencies identified in this analysis could be addressed by the further development of RTI prediction schemes as outlined in section 4. Future research should try to close the large gap between the current state of the practice and the advanced state of the art by proposing incremental and applicable improvements to currently deployed prediction schemes. Furthermore, the evaluation method along with the performance metrics and the analysis of RTI added-value could be applied for assessing the reliability of alternative prediction schemes and thus support a cost-benefit analysis of RTI improvements. In addition, the performance of a certain RTI system could be compared for different lines, modes or even cities. The latter would enable to investigate the relation between service reliability and RTI reliability.

Acknowledgements

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